QRT Formula and Revision Guide

Quant research assessment preparation

Core coverage	Probability, statistics, linear algebra, regression, ML, quant metrics, pandas
Designed for	Fast MCQ recall plus one timed data-science coding task
Use with	`QRT_mock_test.pdf` and `QRT_data_science.ipynb`

How to use this guide

First learn the boxed identities and the meaning of every symbol. Then practise deciding *which* formula or modeling assumption applies. In the final 24 hours, use the Rapid Recall section rather than rereading every derivation.

Scope and evidence

This is an original revision guide, not an official Qube Research & Technologies syllabus and not a collection of leaked questions. It follows the topic mix documented in `sources_and_methodology.md`. The actual invitation and on-screen instructions are authoritative.

Priority map

What to know cold

Priority	Topics	Required depth
Highest	Conditional probability, Bayes, expectation, variance, covariance, z-scores, OLS, regularization, cross-validation, leakage, pandas grouping	Calculate quickly and explain assumptions
High	Hypothesis tests, goodness-of-fit, rank, tree ensembles, bias–variance, class imbalance, feature importance, optimizers	Recognize correct statements and common traps
Medium	PCA, clustering, calibration, drift, time-series diagnostics, portfolio metrics	Know definitions, objectives, and failure modes
Lower	Long derivations, exotic architectures, finance-specific pricing models	Review only if your invitation says so

Recommended 82-minute pacing

Section	Questions	Suggested time
Data-science coding	1	42 minutes
Mathematics and statistics	10	18 minutes
ML fundamentals	5	8 minutes
ML advanced	8	14 minutes

Decision rule under time pressure

For an MCQ, identify the topic, write the smallest relevant identity, eliminate answers that violate units or bounds, and move on. For coding, get a correct end-to-end baseline before polishing; verify row counts, missingness, uniqueness, dates, and output columns.

Notation used throughout

Symbol	Meaning	Symbol	Meaning
X, Y	Random variables	x_i, y_i	Observed values
μ, σ^2	Population mean and variance	$\bar{x}, s^2$	Sample mean and variance
n	Number of observations	p	Number of fitted predictors
$X \in \mathbb{R}^{n \times p}$	Design matrix	β	Coefficient vector
r_t	Return at time t	$s_{i,t}$	Signal for asset i at time t

1. Probability foundations

Conditional reasoning first

Events, complements, and unions

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \qquad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

If A and B are disjoint, $\mathbb{P}(A \cap B) = 0$. Disjoint is not the same as independent.

Conditional probability and Bayes' rule

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \qquad \mathbb{P}(B) > 0,$$

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c)}.$$

More generally, if $A_1, \ldots, A_k$ partition the sample space,

$$\mathbb{P}(B) = \sum_{j=1}^{k} \mathbb{P}(B \mid A_j)\mathbb{P}(A_j).$$

Plain-language interpretation

Bayes updates a prior probability $\mathbb{P}(A)$ using how compatible the evidence B is with A. The denominator must include every way in which B can occur.

Independence

$$A \perp B \iff \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) \iff \mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Zero covariance does not generally imply independence. It does for jointly Gaussian variables.

Expectation, variance, covariance, correlation

For a discrete variable,

$$\mathbb{E}[X] = \sum_{x} x\mathbb{P}(X = x).$$

For a continuous variable with density f,

$$\mathbb{E}[X] = \int x f(x)\, dx.$$

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

$$\operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y], \qquad \operatorname{Corr}(X, Y) = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

Linear transformations

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \qquad \operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X),$$

$$\operatorname{Var}(aX + bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab \operatorname{Cov}(X, Y).$$

If X and Y are independent, the covariance term is zero.

Common probability traps

Do not invert a conditional: $\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$. Do not add probabilities without subtracting the overlap. Remember that variance uses squared coefficients, and the covariance sign follows $2ab$.

2. Expectations and distributions

Recognize the model

Laws of total expectation and variance

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big],$$
$$\mathrm{Var}(X) = \mathbb{E}\big[\mathrm{Var}(X \mid Y)\big] + \mathrm{Var}\big(\mathbb{E}[X \mid Y]\big).$$

The first term is average within-group uncertainty; the second is between-group uncertainty.

Distribution	Support and parameters	Mean	Variance	Typical use
Bernoulli(p)	$x \in \{0,1\}$	p	$p(1-p)$	One binary outcome
Binomial(n,p)	$x = 0,\dots,n$	np	$np(1-p)$	Number of successes
Poisson(λ)	$x = 0,1,\dots$	λ	λ	Event counts per exposure
Uniform(a,b)	$a \le x \le b$	$(a+b)/2$	$(b-a)^2/12$	Bounded equal-density values
Normal(μ,σ^2)	$x \in \mathbb{R}$	μ	σ^2	Additive noise, CLT approximations
Exponential(λ)	$x \ge 0$, rate λ	$1/\lambda$	$1/\lambda^2$	Waiting time, memoryless model
Gamma(α,β)	$x > 0$, shape/rate	α/β	α/β^2	Positive rate or duration
Beta(α,β)	$0 < x < 1$	$\alpha/(\alpha+\beta)$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	Probability parameter

Standardization

If X has mean μ and standard deviation $\sigma > 0$,

$$Z = \frac{X-\mu}{\sigma}.$$

Then $\mathbb{E}[Z] = 0$ and $\mathrm{Var}(Z) = 1$. If $X \sim N(\mu,\sigma^2)$, then $Z \sim N(0,1)$.

Joint density normalization

A valid density satisfies $f(x,y) \ge 0$ and

$$\iint_{\mathcal{S}} f(x,y)\,dx\,dy = 1.$$

If $f(x,y) = c\,g(x,y)$ on support $\mathcal{S}$, then

$$c = \left(\iint_{\mathcal{S}} g(x,y)\,dx\,dy\right)^{-1}.$$

Memoryless property

For an exponential waiting time T, $\mathbb{P}(T > s+t \mid T > s) = \mathbb{P}(T > t)$. The geometric distribution is the discrete analogue.

3. Estimation

From sample to parameter

Sample summaries

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The population-style variance of the observed values uses divisor n; the usual unbiased sample variance uses $n-1$.

Bias, variance, and mean squared error

For estimator $\hat{\theta}$,

$$\mathrm{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta,$$

$$\mathrm{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2.$$

An estimator can accept a small bias to obtain a larger variance reduction.

Method of moments

Equate a theoretical moment to its sample analogue. If $\mathbb{E}_\theta[X] = m(\theta)$, solve

$$m(\hat{\theta}_{\mathrm{MOM}}) = \bar{x}.$$

Example: for $X \sim \mathrm{Uniform}(0, \theta)$, $\mathbb{E}[X] = \theta/2$, so $\hat{\theta} = 2\bar{x}$.

Maximum likelihood and MAP

$$L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta), \qquad \ell(\theta) = \sum_{i=1}^{n} \log f(x_i \mid \theta),$$

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \ell(\theta).$$

With prior $p(\theta)$,

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \{\ell(\theta) + \log p(\theta)\}.$$

The logarithm converts products into sums and preserves the maximizer.

Gamma–Poisson conjugacy

Use a Gamma prior with shape α and rate β:

$$\lambda \sim \mathrm{Gamma}(\alpha, \beta), \qquad x_i \mid \lambda \sim \mathrm{Poisson}(e_i\lambda).$$

Then

$$\lambda \mid x \sim \mathrm{Gamma}\left(\alpha + \sum_i x_i,\ \beta + \sum_i e_i\right),$$

$$\mathbb{E}[\lambda \mid x] = \frac{\alpha + \sum_i x_i}{\beta + \sum_i e_i}.$$

Always check whether the Gamma convention uses rate or scale.

Law of large numbers and central limit theorem

Under standard conditions, $\bar{X}_n \to \mu$ as $n \to \infty$. Approximately,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \dot{\sim} N(0,1).$$

The standard error of the mean is $\sigma/\sqrt{n}$, estimated by $s/\sqrt{n}$.

4. Testing and confidence intervals

Evidence, not certainty

Generic test statistic

$$\text{test statistic} = \frac{\text{estimate} - \text{null value}}{\text{standard error under } H_0}.$$

For a mean with known σ,

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

With unknown σ and suitable normality assumptions,

$$T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \quad \text{under } H_0.$$

Confidence interval

$$\text{estimate} \ \pm \ \text{critical value} \times \text{standard error}.$$

For a large-sample 95% mean interval,

$$\bar{x} \pm 1.96\frac{s}{\sqrt{n}}.$$

The frequentist interval procedure has 95% long-run coverage; the realized fixed parameter is not random.

P-value

The p-value is the probability, assuming the null hypothesis is true, of observing a test statistic at least as incompatible with the null as the one obtained. It is not the probability that the null is true.

Decision	H_0 true	H_0 false
Reject H_0	Type I error, probability α	Correct rejection, power $1-\beta$
Do not reject	Correct non-rejection	Type II error, probability β

Chi-squared goodness-of-fit

For observed counts O_j and expected counts E_j,

$$\chi^2 = \sum_{j=1}^{k} \frac{(O_j - E_j)^2}{E_j}.$$

If q parameters were estimated from the same data, a common reference degree of freedom is

$$\text{df} = k - 1 - q.$$

Expected counts should not be too small for the approximation to be reliable.

Multiple testing

Testing many hypotheses increases false discoveries. Bonferroni controls family-wise error by testing each of m hypotheses at level

$$\alpha_{\text{individual}} = \frac{\alpha}{m}.$$

It is simple and conservative. False-discovery-rate procedures target a different error criterion.

Test traps

Statistical significance is not economic significance. Failing to reject is not proof of equality. Choosing a test after inspecting outcomes invalidates its nominal calibration unless selection is accounted for.

5. Linear algebra

The geometry behind models

Dimensions and matrix products

If $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, then $AB \in \mathbb{R}^{m\times p}$. Generally $AB \neq BA$.

$$(AB)^\top = B^\top A^\top, \qquad (AB)^{-1} = B^{-1}A^{-1} \quad \text{when invertible.}$$

Rank

$\operatorname{rank}(A)$ is the number of linearly independent rows or columns. For $A \in \mathbb{R}^{m\times n}$,

$$\operatorname{rank}(A) \leq \min(m, n).$$

A square matrix is invertible exactly when it has full rank. In regression, full column rank of X gives a unique OLS coefficient vector.

Eigenvalues and positive semidefiniteness

$$Av = \lambda v.$$

A symmetric matrix A is positive semidefinite (PSD) if

$$x^\top Ax \geq 0 \quad \text{for every } x.$$

For symmetric A, PSD is equivalent to all eigenvalues being nonnegative. Covariance matrices are symmetric PSD.

Covariance matrix

For random vector $X \in \mathbb{R}^p$ with mean μ,

$$\Sigma = \mathbb{E}[(X-\mu)(X-\mu)^\top].$$

For any portfolio weights w,

$$\operatorname{Var}(w^\top X) = w^\top \Sigma w \geq 0.$$

Fast rank check

Look first for an all-zero row, proportional rows, or a row that is a sum of others. Then identify a visibly independent subset to establish the exact rank.

6. OLS regression

Formula, assumptions, diagnostics

Simple regression

For $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$,

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{\widehat{\mathrm{Cov}}(X, Y)}{\widehat{\mathrm{Var}}(X)},$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

The fitted line passes through $(\bar{x}, \bar{y})$.

Matrix OLS

$$y = X\beta + \varepsilon, \qquad \hat{\beta} = (X^\top X)^{-1} X^\top y$$

when X has full column rank. OLS minimizes

$$\mathrm{SSE}(\beta) = \|y - X\beta\|_2^2.$$

The residual vector is $e = y - X\hat{\beta}$ and satisfies $X^\top e = 0$.

6. OLS regression continued

Know what each assumption buys

Condition	Main implication	If violated
Correct linear conditional mean	Model form is appropriate	Systematic residual structure, biased predictions
$\mathbb{E}[\varepsilon \mid X] = 0$	Conditional unbiasedness of OLS	Endogeneity and biased coefficients
Full column rank of X	Unique OLS solution	Perfect multicollinearity, non-identification
Homoskedastic uncorrelated errors	Classical OLS variance formula, efficiency among linear unbiased estimators	Standard errors need robust or structured correction
Normal errors	Exact small-sample t/F inference	Often not needed asymptotically or for unbiasedness

Fit metrics

$$\mathrm{SST} = \sum_i (y_i - \bar{y})^2, \qquad \mathrm{SSE} = \sum_i (y_i - \hat{y}_i)^2,$$

$$R^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}},$$

$$\bar{R}^2 = 1 - (1 - R^2)\frac{n-1}{n-p-1},$$

where p excludes the intercept. Adding predictors cannot decrease training R^2, but it can decrease adjusted R^2 and out-of-sample performance.

Ridge and Lasso

Ridge:

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta} \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right\}.$$

Lasso:

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta} \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\}.$$

Increasing λ generally increases bias and reduces variance. Ridge smoothly shrinks correlated coefficients; Lasso can set coefficients exactly to zero. Scale numeric features before penalization. The intercept is usually not penalized.

Regression traps

Homoskedasticity and normality are not required for OLS coefficients to be unbiased. Correlation does not imply causation. A high in-sample R^2 does not establish stable out-of-sample prediction.

7. Machine-learning workflow

Evaluation is part of the model

Split roles

The training set fits parameters. The validation set or inner CV selects models and hyperparameters. The test or outer holdout estimates final generalization. Reusing the holdout for decisions turns it into training data.

Empirical risk minimization

Given loss ℓ and predictor f_θ,

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} \ell(y_i, f_\theta(x_i)) + \lambda \Omega(\theta).$$

The regularizer Ω encodes a preference such as small weights or sparsity.

Bias–variance decomposition under squared loss

At a fixed input x, with irreducible noise variance σ^2,

$$\mathbb{E}[(Y - \hat{f}(x))^2] = \text{Bias}(\hat{f}(x))^2 + \text{Var}(\hat{f}(x)) + \sigma^2.$$

More flexible models usually reduce bias but may increase variance. Regularization, more data, bagging, and simpler models can reduce variance.

Cross-validation

- In ordinary K-fold CV, train on $K-1$ folds and validate on the remaining fold; average the validation metric.
- Use stratification for rare classes when random splitting is appropriate.
- For time series, preserve chronology: expanding or rolling training windows followed by later validation blocks.
- Fit preprocessing separately inside each training fold: imputation, scaling, PCA, feature selection, and resampling.
- Nested CV uses inner folds for selection and outer folds to estimate the whole selection procedure.

Leakage patterns

Leakage includes fitting a scaler or PCA on all rows, imputing with future values, using a feature recorded after the target horizon, oversampling before splitting, selecting features with the full dataset, or randomly splitting dependent time-series observations.

Preprocessing facts

Operation	Important for	Usually less important for
Scaling	Linear penalties, KNN, SVM, PCA, neural nets	Decision trees and tree ensembles
Imputation	Models that reject missing values	Models with explicit missing-value handling
One-hot encoding	Unordered categories in linear/distance models	Native categorical algorithms
Winsorization	Limiting extreme-value influence	Should not replace understanding data errors

8. Predictive metrics

Choose the metric for the decision

Regression

MSE, RMSE, MAE, and R^2

$$\mathrm{MSE} = \frac{1}{n}\sum_i (y_i - \hat{y}_i)^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}},$$

$$\mathrm{MAE} = \frac{1}{n}\sum_i |y_i - \hat{y}_i|, \qquad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

MSE penalizes large errors more strongly; MAE is more robust to outliers. Test R^2 can be negative.

Binary classification

	Predicted positive	Predicted negative
Actually positive	TP	FN
Actually negative	FP	TN

$$\mathrm{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\mathrm{Specificity} = \frac{TN}{TN + FP}$$

$$\mathrm{Precision} = \frac{TP}{TP + FP}$$

$$F_1 = 2\frac{\mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$$

$$\mathrm{Recall} = \frac{TP}{TP + FN}$$

$$\mathrm{FPR} = \frac{FP}{FP + TN}$$

Binary log loss

For predicted probabilities p_i,

$$\mathrm{LogLoss} = -\frac{1}{n}\sum_i \left[y_i \log p_i + (1 - y_i)\log(1 - p_i)\right].$$

It rewards calibrated probabilities and heavily penalizes confident wrong predictions.

ROC-AUC, PR-AUC, and calibration

ROC-AUC is a ranking measure: it is the probability that a randomly chosen positive receives a higher score than a randomly chosen negative. A strictly increasing score transformation preserves AUC but can change calibration and log loss. With rare positives, precision–recall curves are often more informative than accuracy or ROC-AUC alone.

Imbalance

A classifier predicting the majority class can have high accuracy and zero usefulness. Consider PR-AUC, recall at a precision constraint, class weights, threshold selection, and resampling performed only within training folds.

9. Trees and ensembles

How the methods differ

Classification impurity

For class proportions $p_1, \dots, p_K$,

$$\text{Gini} = 1 - \sum_{k=1}^{K} p_k^2, \qquad \text{Entropy} = -\sum_{k=1}^{K} p_k \log p_k.$$

A split is selected to maximize impurity reduction, equivalently the parent impurity minus the weighted child impurities.

Method	Main mechanism	Typical strength	Main risk
Decision tree	Recursive feature splits	Nonlinear interactions, little preprocessing	High variance and overfitting
Bagging	Average models fitted on bootstrap samples	Variance reduction	Limited bias reduction
Random forest	Bagging plus random feature subsets at splits	Decorrelated strong baseline	Large ensembles, correlated-feature interpretation
Gradient boosting	Sequentially fit weak learners to negative gradients	Strong predictive accuracy, bias reduction	Sensitive tuning and overfitting

Gradient boosting update

For loss L and current model F_{m-1}, pseudo-residuals are

$$r_{im} = -\left.\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)}\right|_{F=F_{m-1}}.$$

Fit weak learner h_m to these targets and update

$$F_m(x) = F_{m-1}(x) + \eta h_m(x),$$

where η is the learning rate. Under squared error, pseudo-residuals are ordinary residuals.

Random forest decorrelation

Bootstrap rows make training samples differ; random feature subsets make split opportunities differ. Averaging helps most when individual trees are accurate but not too correlated.

Feature importance

- Impurity importance can favor high-cardinality or frequently split features.

- Permutation importance measures the performance loss after disrupting one feature.
- With correlated predictors, permuting one may show little loss because another substitutes for it.
- Grouped permutation, drop-column tests, and stability across folds provide useful sensitivity checks.
- SHAP or local explanations describe a fitted model, not causal effects.

10. PCA, clustering, and neural networks

Advanced fundamentals

Principal component analysis

Center the feature matrix and compute covariance S. The first principal direction solves

$$v_1 = \arg\max_{\|v\|_2=1} v^\top S v.$$

It is the eigenvector with largest eigenvalue λ_1. The explained-variance ratio of component j is

$$\frac{\lambda_j}{\sum_k \lambda_k}.$$

PCA is unsupervised, sensitive to feature scale, and must be fitted only on training data within each fold.

K-means objective

For clusters $C_1, \dots, C_K$ with centroids μ_k,

$$\min_{C_1,\dots,C_K} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2.$$

K-means favors roughly spherical, similarly scaled clusters and can converge to local optima; initialize multiple times.

Common activation and classification formulas

$$\mathrm{ReLU}(z) = \max(0, z), \qquad \sigma(z) = \frac{1}{1+e^{-z}},$$

$$\mathrm{softmax}(z)_k = \frac{e^{z_k}}{\sum_j e^{z_j}}.$$

For one-hot target y and predicted class probabilities p,

$$\mathrm{CrossEntropy} = -\sum_k y_k \log p_k.$$

Gradient descent, momentum, and Adam

Plain gradient descent:

$$\theta_{t+1} = \theta_t - \eta g_t.$$

Momentum keeps an exponential direction estimate. Adam keeps first and second moment estimates:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t,$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2,$$

then uses bias-corrected $\hat{m}_t, \hat{v}_t$:

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}.$$

Adam provides adaptive parameter scaling; it does not remove learning-rate tuning or guarantee a global optimum.

Regularizing neural networks

Weight decay, dropout, early stopping, data augmentation, smaller architectures, and more data can reduce overfitting. Training loss alone cannot select the stopping point without optimistic bias.

11. Drift and model selection

What changes after deployment

Change	Mathematical description	Example
Covariate shift	$P(X)$ changes, $P(Y \mid X)$ stable	Different feature mix
Label shift	$P(Y)$ changes, $P(X \mid Y)$ stable	Base event rate changes
Concept drift	$P(Y \mid X)$ changes	Old signal–return relation weakens

Monitoring

Monitor input distributions, missingness, score distributions, calibration, ranking metrics, realized outcomes, latency, and subgroup behavior. Stable feature marginals do not rule out concept drift.

Nested cross-validation

For each outer fold: reserve the outer validation data; use only the outer training portion to run inner hyperparameter selection; refit the selected specification on the outer training portion; score once on the outer validation portion. Average outer scores to estimate the complete selection procedure.

Hyperparameter tuning trap

Reporting the best score among many configurations on the same validation set is optimistically biased. The more configurations tried, the stronger the selection effect.

12. Quant and time-series formulas

Signal quality is not a backtest

Simple and log returns

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}, \qquad r_t = \log \frac{P_t}{P_{t-1}} = \log(1 + R_t).$$

Simple returns compound multiplicatively:

$$1 + R_{1:T} = \prod_{t=1}^{T} (1 + R_t).$$

Log returns add:

$$r_{1:T} = \sum_{t=1}^{T} r_t.$$

For small returns, $\log(1 + R) \approx R$.

Annualization

With approximately 252 trading days per year,

$$\widehat{R}_{\text{ann}} = 252\bar{r}, \qquad \widehat{\sigma}_{\text{ann}} = \sqrt{252}\, s_r.$$

With zero risk-free rate,

$$\widehat{\text{Sharpe}} = \frac{\widehat{R}_{\text{ann}}}{\widehat{\sigma}_{\text{ann}}} = \sqrt{252}\frac{\bar{r}}{s_r}.$$

These formulas assume the daily sampling and dependence structure make square-root-of-time scaling reasonable.

12. Quant formulas continued

Cross-sectional evaluation

Let I_t be available assets, $n_t = |I_t|$, and

$$k_t = \max\left(1, \left\lfloor \frac{n_t}{5} \right\rfloor\right).$$

Let L_t be the k_t highest-signal assets and S_t the k_t lowest-signal assets.

Long–short factor return

$$R_t^{\text{factor}} = \frac{1}{k_t}\sum_{i\in L_t} R_{i,t+1} - \frac{1}{k_t}\sum_{i\in S_t} R_{i,t+1}.$$

Positive return means high-signal assets outperformed low-signal assets that day. This ignores costs unless they are explicitly deducted.

Information coefficient

Pearson IC:

$$\text{IC}_t = \text{Corr}_i(s_{i,t}, R_{i,t+1}).$$

Rank IC:

$$\text{RankIC}_t = Corr_i(\text{rank}(s_{i,t}), \text{rank}(R_{i,t+1})).$$

Rank IC measures monotonic ordering and is less sensitive to extreme magnitudes. Use average ranks for ties.

Equity curve and drawdown

$$E_t = \prod_{j=1}^{t}(1+R_j), \qquad D_t = \frac{E_t}{\max_{u\le t} E_u} - 1,$$

$$\text{MaxDrawdown} = \min_t D_t.$$

Drawdown is normally nonpositive; -0.08 means an 8% decline from a previous peak.

Turnover and net return

One common one-way turnover definition is

$$\text{Turnover}_t = \frac{1}{2}\sum_i |w_{i,t} - w_{i,t-1}|.$$

With proportional cost c,

$$R_t^{\text{net}} \approx R_t^{\text{gross}} - c\,\text{Turnover}_t.$$

Always state the turnover convention and whether weights are measured before or after market movement.

Autocorrelation

At lag k,

$$\rho_k = \mathrm{Corr}(X_t, X_{t-k}).$$

Serial dependence reduces the amount of independent information and can make naive standard errors and Sharpe interpretations too optimistic.

Backtest failure modes

Look-ahead bias, survivorship bias, data revisions, timestamp misalignment, overlapping labels, multiple testing, unrealistic fills, omitted costs, shorting constraints, regime instability, and using the same holdout repeatedly can all create false confidence.

13. Pandas coding reference

Operations to know under a clock

Load, inspect, and enforce types

```
df = pd.read_csv("data.csv")
df.shape
df.head()
df.info()
df.isna().sum()
df["date"] = pd.to_datetime(df["date"])
df["value"] = pd.to_numeric(df["value"], errors="coerce")
```

Filter, sort, and deduplicate

```
start = pd.Timestamp("2025-06-02")
holdout = df.loc[df["date"] >= start].copy()

df = (
    df.sort_values(["date", "asset", "observation_time"],
                   kind="mergesort")
      .drop_duplicates(["date", "asset"], keep="last")
      .reset_index(drop=True)
)
```

Stable sorting makes deterministic tie handling easier. After sorting corrections from early to late, `keep="last"` keeps the latest correction.

GroupBy: aggregate versus transform

```
# One output row per date:
daily_mean = df.groupby("date")["value"].agg("mean")

# One output value per original row:
row_aligned_mean = df.groupby("date")["value"].transform("mean")

# Same-size transformation within each date:
df["z"] = df.groupby("date")["value"].transform(
    lambda s: (s - s.mean()) / s.std(ddof=0)
)
```

Transform mental model

Pandas passes each group Series to the function. If the function returns one scalar, `transform` broadcasts it to every row in that group. If it returns a same-length Series, pandas aligns those values back to the original rows.

Missing values, quantiles, and clipping

```
sector_median = (
    df.groupby(["date", "sector"])["signal_raw"]
      .transform("median")
)
day_median = df.groupby("date")["signal_raw"].transform("median")
df["signal_raw"] = df["signal_raw"].fillna(sector_median)
df["signal_raw"] = df["signal_raw"].fillna(day_median)

lower = df.groupby("date")["signal_raw"].transform(
    lambda s: s.quantile(0.05)
)
upper = df.groupby("date")["signal_raw"].transform(
    lambda s: s.quantile(0.95)
)
df["signal_winsor"] = df["signal_raw"].clip(lower, upper)
```

13. Pandas continued

Daily portfolios and time series

Iterate over daily cross-sections

```
rows = []
for date, group in holdout.groupby("date", sort=True):
    group = group.sort_values(["signal_z", "asset"],
                              kind="mergesort")
    n = len(group)
    k = max(1, n // 5)
    short = group.head(k)
    long = group.tail(k)

    signal_rank = group["signal_z"].rank(method="average")
    return_rank = group["next_day_return"].rank(method="average")

    rows.append({
        "date": date,
        "factor_return": (long["next_day_return"].mean()
                          - short["next_day_return"].mean()),
        "rank_ic": signal_rank.corr(return_rank),
        "n_assets": n,
        "n_per_leg": k,
    })

daily = pd.DataFrame(rows).sort_values("date").reset_index(drop=True)
```

Time-series operations

```
df = df.sort_values(["asset", "date"])
df["lag_signal"] = df.groupby("asset")["signal"].shift(1)
df["return"] = df.groupby("asset")["price"].pct_change()
```

```
df["rolling_mean"] = (
    df.groupby("asset")["return"]
      .transform(lambda s: s.rolling(20, min_periods=20).mean())
)
```

Join and reshape

```
merged = left.merge(right, on=["date", "asset"],
                    how="left", validate="one_to_one")
stacked = pd.concat([df_a, df_b], ignore_index=True)
wide = df.pivot(index="date", columns="asset", values="return")
```

Use `validate=` in merges when you know the expected key relationship. It catches accidental many-to-many joins.

Coding traps

Do not overwrite the raw input unexpectedly. Avoid chained assignment; filter with `.loc[...].copy()`. Sort before `head`, `tail`, `shift`, rolling calculations, or deduplication. Verify that every feature is available at prediction time.

14. Data-science validation checklist

Test invariants, not just values

After cleaning

- Expected number of rows
- No duplicate key pairs
- Correct dtypes
- Missingness understood
- Dates ordered
- Original raw data not mutated

After feature construction

- Daily z-score means near zero
- Daily population standard deviations near one
- No target used in signal construction
- Lag directions correct
- Train/holdout boundary respected
- Tie handling deterministic

```
assert not clean.duplicated(["date", "asset"]).any()
assert clean["signal_z"].notna().all()

by_day = clean.groupby("date")["signal_z"]
assert np.allclose(by_day.mean(), 0.0, atol=1e-10)
assert np.allclose(by_day.std(ddof=0), 1.0, atol=1e-10)

assert daily["date"].is_monotonic_increasing
assert daily[["factor_return", "rank_ic"]].notna().all().all()
```

Good implementation order

Parse and validate inputs; build the smallest correct transformation; inspect shape and keys; add the portfolio calculation; add metrics; write assertions; restart the kernel and run top to bottom.

15. Common conceptual traps

Tempting answers to reject

1. **Conditional probability:** the base rate belongs in Bayes' numerator and denominator.
2. **Variance:** constants shift the mean but do not change variance; scaling by a scales variance by a^2.
3. **OLS:** homoskedasticity is not required for unbiasedness; zero conditional mean is.

4. **Regularization:** stronger penalties generally increase bias and reduce variance; Ridge does not usually create exact zeros.
5. **Cross-validation:** preprocessing belongs inside each training fold.
6. **Time series:** random splitting can leak future regimes and neighboring observations.
7. **AUC:** ranking quality is not probability calibration.
8. **Feature importance:** predictive attribution is not causal effect; correlation can hide importance.
9. **Backtests:** a high Sharpe without costs, uncertainty, or an untouched holdout is not enough.
10. **Pandas:** `agg` reduces groups; `transform` returns values aligned to original rows.

16. Rapid Recall

Final review sheet

Topic	Formula or fact
Conditional probability	$\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$
Bayes	Posterior $\propto$ likelihood $\times$ prior
Total probability	$\mathbb{P}(B) = \sum_j \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)$
Variance identity	$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$
Linear variance	$\operatorname{Var}(aX + bY) = a^2 \operatorname{Var} X + b^2 \operatorname{Var} Y + 2ab \operatorname{Cov}(X, Y)$
Standard score	$z = (x - \mu)/\sigma$
Sample variance	$s^2 = \sum_i (x_i - \bar{x})^2/(n-1)$
Estimator MSE	$\text{MSE} = \text{Variance} + \text{Bias}^2$
Method of moments	Equate theoretical and empirical moments
MLE	Maximize $\sum_i \log f(x_i \mid \theta)$
Gamma–Poisson	Posterior shape $= \alpha + \sum x_i$; rate $= \beta + \sum e_i$
Standard error	$s/\sqrt{n}$ for the sample mean
Confidence interval	Estimate $\pm$ critical value $\times$ SE
Goodness-of-fit df	Categories $-1-$ fitted parameters
Rank	Number of independent rows or columns
PSD	$x^\top Ax \geq 0$ for all x; symmetric PSD matrices have nonnegative eigenvalues
OLS slope	$\hat{\beta}_1 = \widehat{\operatorname{Cov}}(X, Y)/\widehat{\operatorname{Var}}(X)$
OLS intercept	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
Matrix OLS	$\hat{\beta} = (X^\top X)^{-1} X^\top y$ under full column rank
R^2	$1 - \text{SSE}/\text{SST}$
Ridge / Lasso	L_2 smooth shrinkage / L_1 possible exact zeros
Bias–variance	Error $= \text{bias}^2 +$ variance $+$ irreducible noise
Precision / recall	$TP/(TP + FP)$ and $TP/(TP + FN)$
F_1	Harmonic mean of precision and recall
Log loss	$-[y \log p + (1 - y) \log(1 - p)]$ averaged over rows
Random forest	Bootstrap rows plus random feature subsets; average trees
Boosting	Fit negative gradients sequentially; update with learning rate
PCA	Eigenvectors of covariance; largest eigenvalue gives first component
K-means	Minimize within-cluster squared distances
Adam	Moving first and second gradient moments with adaptive scaling
Simple return	$R_t = P_t/P_{t-1} - 1$
Compounding	$1 + R_{1:T} = \prod_t (1 + R_t)$
Annualized return / vol	$252\bar{r}$ and $\sqrt{252}s_r$
Sharpe	$\sqrt{252}\bar{r}/s_r$ when risk-free rate is zero
Rank IC	Correlation of cross-sectional signal and future-return ranks
Drawdown	$E_t/\max_{u \leq t} E_u - 1$
Pandas transform	Same-indexed output; scalar group results are broadcast

Last-minute rule

Do not memorize a formula without its conditions. Ask: What is random? What is conditioned on? Is the statistic cross-sectional or time-series? Was every transformation fitted without using future or holdout information?

17. Final checklist

Before pressing Start

Mathematics

- Bayes and total probability
- Expectation, variance, covariance
- Common distribution means and variances
- MOM, MLE, posterior update
- z/t statistics and chi-squared df
- Matrix rank and covariance PSD
- OLS slope, intercept, assumptions
- Ridge versus Lasso

Machine learning

- Bias–variance tradeoff
- Leakage-safe cross-validation
- Class imbalance metrics
- Bagging versus boosting
- Random forest decorrelation
- PCA and scaling
- Feature importance caveats
- Adam and learning-rate facts

Coding

- Parse dates and sort explicitly
- Deduplicate with a clear rule
- Use groupwise `transform`
- Handle missing values and zero variance
- Use deterministic tie-breaking
- Respect the holdout boundary
- Return exact requested columns
- Restart and run all cells

Research judgement

- Prediction timestamp precedes target
- Costs and turnover considered
- Holdout untouched during selection
- Multiple testing acknowledged
- Performance uncertainty reported
- Regime and drift risks considered
- Economic and statistical significance separated

During the test

Read whether a question is single-answer or select-all. For formulas, write the structure before substituting numbers. For code, preserve the raw data, name intermediate objects clearly, and validate invariants. If stuck, leave a working partial implementation and move forward.

Good research is correct, reproducible, leakage-aware, and skeptical of its own backtest.